

The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate

A Trisductive Structural Framework Whose Empirical Content Is Geometric Necessity and Whose One Free Input Is an Unmeasured Substrate Scale

Mohammad F. Islam¹

¹Independent Theoretical Research, United States. Correspondence: islammm@alummi.iu.edu.

The Standard Model and general relativity describe the four interactions and the particle spectrum with extraordinary precision, yet derive almost none of their structure: the gauge group, the force hierarchy, the spin content of each mediator, and the dark sector all enter as measured inputs rather than as consequences. The missing element is not computational power but a common substrate whose geometry forces the structure, and the leading extensions add inputs, extra dimensions, superpartners, or imposed discrete symmetries, rather than removing them. This paper develops a single actuating continuous substrate, read through a three-way orthogonal decomposition into a formal, a thermodynamic, and a registrational warrant, on which each force is a distinct configuration classified by the geometry of its source tensor and each particle is a topological knot, with geometry fixed before magnitude. The framework's empirical content is dominated by geometric necessity rather than fitted number: gravitational radiation is purely quadrupolar, magnetic monopoles are absent, color is exactly confined, and the neutrino is a Majorana particle, each forced by a topology or a source structure; the normal neutrino ordering and the proton's effective stability follow as structural consequences; the three-generation count is inherited and read through an open conjecture rather than derived; and the dark-energy equation of state stays at or above minus one with a small present departure, the one prediction currently disfavored by baryon-acoustic-oscillation data, which prefer a past excursion below minus one that the framework forbids. Each is falsifiable at five standard deviations by a named instrument. If confirmed, the account unifies by placing the interactions on one substrate rather than by computing their magnitudes, and it states in advance the single unmeasured input, the thermodynamic characterization of the substrate, in which every remaining continuous constant resides, alongside a small closed set of discrete structural stipulations it marks rather than derives. The paper claims structural placement and geometric prediction; it does not claim to derive the constants, and it declines the numerical coincidences that would appear to.

KEYWORDS topological field theory, gauge structure, spin-2 exhaustion, Hodge decomposition, color confinement, neutrino mass ordering, dark energy equation of state, geometric necessity

1. Background: the gaps are structural, not computational

The Standard Model is the most precisely tested theory in the history of physics, and general relativity has passed every test from the perihelion of Mercury to the direct detection of gravitational waves²⁴. Neither is under challenge here. What is under challenge is a specific and rarely stated fact: both theories

describe their objects without deriving the structure of those objects. The distinction is between a barrier that better computation will remove and a barrier that requires a different geometry, and the second kind is the subject of this paper.

Five structural gaps make the point precisely, and each is a place where the standard formalism contains no invariant that would force the observed structure. The first is the gauge group. The product structure of the strong, weak, and electromagnetic

interactions is an empirical input; no principle inside the Standard Model selects it over a larger or smaller product, and the number of colors is observed rather than derived. The second is the force hierarchy. The ratio between the strong coupling and the gravitational coupling at the scale of the proton spans roughly thirty-eight orders of magnitude, and the couplings enter as independent measured numbers with no shared origin. The third is the spin content. That the graviton carries spin two, the gauge bosons spin one, and no fundamental scalar force is observed, is read off experiment and imposed, not deduced from the character of each source. The fourth is the dark sector. The rotation curves of galaxies and the accelerating expansion of the universe are described by an added mass component and an added energy component whose microphysics is unknown, and the standard account supplies parameters rather than structure. The fifth is the vacuum. The energy density that quantum field theory assigns to the ground state exceeds the measured cosmological term by many tens of orders of magnitude, the most severe quantitative mismatch in physics, and no mechanism inside the standard framework removes it.

Each of these is structural in the exact sense that increasing precision does not close it. A more accurate lattice computation of the hadron spectrum will not produce a theorem that the perturbative formalism lacks; a more sensitive collider will not derive the gauge group it measures. The barrier is that the formalism carries no topological or geometric invariant whose conservation or exhaustion would force the observed structure. Where such an invariant exists, the structure follows without a free parameter, and the burden of this paper is to exhibit, force by force and sector by sector, where an invariant is available and where instead only an unmeasured magnitude remains.

The stance taken here is therefore modest in one direction and strict in another. It is modest because it does not attempt to compute the constants of nature; the framework will be shown to donate no numerical value to any coupling, mass, or scale. It is strict because it insists that every claim it does make be a geometric necessity, a statement forced by the topology or the source structure of a configuration, falsifiable by a definite measurement, and typed at an honest grade. A structural claim is labeled structural, a claim that follows as a theorem from stated premises is labeled a theorem, a premise is labeled a premise, and a result carried unchanged from established physics is labeled inherited and cited. This grading travels with every claim in the paper and is the discipline that separates a placement from an overclaim.

2. Prior approaches and their common input

Five programs extend the Standard Model toward unification, and each is reviewed against the five gaps of Section 1. Each is given its due, and each is then examined for the input it must add.

lead. String theory naturally contains a spin-two excitation and achieves ultraviolet finiteness, and its correspondence with lower-

dimensional gauge theory has wide application. Its structural cost is the compactification landscape: an enormous number of vacua, each with a different gauge group and different couplings, so that the framework accommodates the Standard Model without selecting it. The landscape is the absence of a prediction rather than the presence of one.

Loop quantum gravity supplies discrete spectra for area and volume and reproduces black-hole entropy from a counting of states. Its cost is that the matter content is not derived from the gravitational quantization; the gauge group, the force hierarchy, and the cosmological term are left untouched.

Grand unified theories reduce the three gauge factors to one and correctly predict the weak mixing angle from a normalization condition run down in energy¹⁸. Their cost is that the unifying group is chosen rather than derived, and the generation structure, the fermion mass hierarchy, and the cosmological term remain external.

Supersymmetry stabilizes the hierarchy between the electroweak and the Planck scale by a cancellation between particles and their partners. No partner has been observed through the third run of the Large Hadron Collider, and even were one found, the mechanism resolves the hierarchy by cancellation rather than by structural derivation.

Extra-dimensional models generate a geometric hierarchy, but the topology and size of the extra dimensions are free parameters, and the gauge group is inserted rather than derived. Across all five programs a single structural pattern recurs: each closes a gap by adding an input, a landscape, a chosen group, a partner spectrum, a compactification, rather than by supplying an invariant that removes the input. The framework of this paper is built on the opposite discipline. It adds no dimensions, no partners, and no imposed symmetry; it places the known structure on one substrate and reports, honestly, exactly where an invariant is available and where an unmeasured magnitude is all that remains.

3. Method: three orthogonal warrants and geometry before magnitude

The method rests on three requirements applied to every structural claim, and on one ordering constraint that fixes the sequence in which the requirements are read.

The first requirement is a three-way orthogonal decomposition, here called the triaxial decomposition. By this is meant the resolution of any claim about a physical configuration into three mutually independent warrants: a formal-structural warrant carrying the topology and the field content, a thermodynamic warrant carrying the energetic and directional content, and a registrational warrant carrying what a measurement records. The three are required to be genuinely independent, so that agreement among them is evidence and a collapse of any one is a definite failure rather than a soft anomaly. The decomposition is not a metaphor; for a system of field configurations it is realized as

three orthogonal reading roads whose mutual independence is checkable, and its use in this paper is to force every sector to be described formally, energetically, and observationally at once.

The second requirement is a three-state verdict economy. By this is meant the restriction of every structural verdict to exactly three values: sealed, meaning the three warrants agree and the structure holds; broken, meaning a named warrant fails and the structure does not hold; and under-determined, meaning the warrants are consistent but insufficient to decide. Fractional and probabilistic truth values are not admitted at the structural level. The economy exists to prevent a reframe from moving a verdict: a verdict changes only on new structural argument or new mathematical content, never on a change of emphasis, and consensus carries no evidential weight in either direction.

The third requirement is the independence criterion for empirical tests. Every prediction must be testable by more than one decentralized instrument using an orthogonal measurement modality, so that no single facility or shared calibration can monopolize the framework's falsification. This is a safeguard against theoretical echo chambers and against a data monoculture read as a structure fact.

lead. The ordering constraint is the load-bearing part of the method and deserves a name of its own: geometry before magnitude. The formal and geometric warrants are read first and fix what a configuration is, its topology, its source structure, its symmetry, and the numerical warrant is read last and reports only magnitude on a structure it did not author. A consequence follows that governs the entire paper: a magnitude that is manufactured by a choice of chart or fitting scheme, and that carries no information about the underlying structure, may not enter a verdict as a structure fact. This single rule is what disqualifies the numerical coincidences that unification programs are often built on, and the paper will invoke it explicitly at the one place, the dark sector, where such a coincidence is available and must be declined.

4. The proposed structure: one substrate, read three ways, and its sectors

4.1 The actuating substrate and its ground state

Formalization.. The framework posits one continuous substrate on which every existent carries positive kinetic content. In physical terms, no configuration that exists is at exact rest at the level of its constituents: the kinetic-energy bound that follows from the canonical commutation relation is strictly positive, the zero-point energy of any bound mode is strictly positive, and the rigorous third law forbids a state cooled to exact zero²⁶. This actuation floor is inherited theorem-grade physics for confined quantum existents; its extension to a claim about every existent whatever is a premise, labeled as such, and the framework earns

its weight from the theorem-grade content downstream, never from inflating the premise. The substrate is taken to be monist in the specific sense that its material content, its topology, and its actuation are three readings of one object rather than three separate things, a structural posit consistent with the continuous-field formulation of quantum field theory and with the absence of any observed separation among the three.

A corollary fixes the character of the vacuum. The ground state of the substrate is not the mathematical void of vanishing field but an isometrically balanced state: the vector sum of its internal components is zero while the magnitude of each component is nonzero. Vector cancellation is not scalar absence, and the distinction is forced by the zero-point energy, the Casimir pressure of the vacuum, and the third law. The vacuum is therefore full and balanced, which is the structural basis, taken up in Section 4.8, for reading the cosmological term as a residual of a balanced state rather than as the energy of an empty one.

4.2 The one decomposition, and why three axes

Formalization.. Every configuration on the substrate is read through the triaxial decomposition of Section 3, and the count of exactly three axes is not chosen but forced. Two independent arguments fix it. Semantically, a complete factual claim about an actuating existent resolves under a deletion test into exactly three irreducible slots, an existence slot, a kinetic slot, and a relational slot, with no fourth slot surviving and no slot removable; this is an operational and reproducible result, and no uniqueness theorem of first-order logic is claimed for it. Algebraically, a verification structure closed under its own composition, associative, without zero divisors, and carrying more than one independent axis, completes uniquely to the quaternions by the classification of the real division algebras⁵, in which the imaginary part is exactly three-dimensional and the real line is the center. The next admissible dimension is eight, and the octonions fail associativity, so a fourth independent verification axis cannot close^{6,7}. The two arguments share no premise, and their agreement on the number three is the operational meaning of the decomposition being real rather than stipulated.

The decomposition carries a directional content that the magnitude alone cannot. The squared magnitude of the closed three-axis structure is invariant under reflection of any single axis, so at the level of magnitude a claim and its negation are indistinguishable; the direction is carried by the axes, the ordered formal decomposition, the handedness of the configuration, and the thermodynamic arrow, and is read from them and never from the bare magnitude. This is the formal ground of geometry before magnitude, and it recurs in every sector below as the rule that the sign and the sense of a structure are read from its geometry, not from its scalar.

Table 1 | The four interactions as configurations of one substrate, classified by source-tensor geometry. The classification is structural; the field equations and constants in each row are inherited from established physics and cited in the text. Grade: structural on the typing, inherited on the content.

Interaction	Configuration type	Source geometry	Range	Mediator spin	Content grade
Gravity	rank-2 tensor field	mass-energy, rank two	infinite, inverse square	two	general relativity, inherited
Electromagnetism	closed two-form	charge-current	infinite, inverse square	one	Maxwell, inherited
Weak	screened chiral vector	left-handed doublet	short, Yukawa	one, massive	electroweak breaking, inherited
Strong	internal braid closure	three-strand winding	sub-femtometer	one, eight states	confinement, inherited

4.3 Gravity: the rank-2 tensor sector

Formalization.. Gravity is read as a rank-two tensor field on the substrate, sourced by the rank-two stress-energy tensor, whose static weak-field limit is the position-space gradient of a mass-energy potential. The configuration type is the tensor field, and the scalar potential is only its static reduction. The spin of the mediator is not assumed but exhausted: a massless field sourced by a symmetric rank-two tensor, expanded around flat space and required to be gauge-consistent, carries exactly the two transverse traceless polarizations of a spin-two field, the component count reducing from ten to two after the four gauge and four constraint conditions are removed, and the linear theory is the Fierz-Pauli theory whose unique consistent completion is general relativity^{1,2}. The field equations are therefore inherited rather than reinvented, and they may be reached from the substrate side as a thermodynamic identity: imposing the proportionality of heat and horizon-area change across every local causal horizon yields the Einstein equation as an equation of state^{3,4}. In the static limit the potential satisfies the flat-space Laplace equation away from sources, so a mass is a sink of the same field for which diffusion is a source, an exact structural mirror that breaks only at the radiative seam where the elliptic static problem becomes the hyperbolic wave problem.

The gap it fills.. General relativity takes the spin-two character of the graviton and the tensor structure of its source as inputs justified after the fact. The gap is the absence of a reason that the mediator must be spin two and cannot be spin zero or spin one. The exhaustion argument supplies that reason as a counting theorem: given a rank-two source, no other spin is available to a consistent massless mediator. The content is inherited and the reason is theorem-grade; what the framework adds is the placement of that reason as a necessity rather than a convention.

Prediction, a geometric necessity.. Because the source is rank two and the mediator is spin two by exhaustion, gravitational radiation carries only the two quadrupolar polarizations and no scalar breathing mode and no vector mode. This is a structural necessity, not a fitted number, and its detection or refutation is direct, as developed in Section 5.

4.4 Electromagnetism: the closed two-form sector

Formalization.. Electromagnetism is read as the exact and harmonic parts of a closed two-form on the substrate. On a spatial slice the electromagnetic two-form decomposes, by the Hodge theorem for square-integrable forms, into mutually orthogonal exact, co-exact, and harmonic parts⁸. Imposing closure, that the two-form has vanishing exterior derivative, is the substrate-level statement of the Bianchi identity, and it eliminates the co-exact part exactly: a form that is both closed and co-exact is harmonic and orthogonal to the image of the codifferential, hence zero. The two-form reduces to an exact part, which is standard electromagnetism with its gauge freedom the freedom of the closed-form representation, and a harmonic part, which carries topological flux read through the Aharonov-Bohm phase by the correspondence between harmonic representatives and cohomology classes⁹. The electric and magnetic fields are not two objects but two projections of this one two-form; a change of inertial frame rotates the electric projection into the magnetic projection, so they are one sector read in a frame, and separating them into two independent forces would misread a frame choice as a structure. The relation between the two response coefficients and the speed of light is a wave-operator property of the field, consistent with the two-sector decomposition rather than derived from it; the numerical value of the coefficients does not follow structurally, and is treated as an unmeasured input in Section 4.10.

The gap it fills.. Standard electromagnetism treats the absence of magnetic monopoles as an empirical fact and the closure of the field as a convention of the bundle description. The gap is the absence of a structural statement that forbids the monopole. Closure supplies it: the monopole is exactly the co-exact part that closure removes, so its absence is a structural consequence rather than an accident, contingent only on closure holding wherever it is tested¹⁰.

Prediction, a geometric necessity.. Because closure removes the co-exact sector, no magnetic monopole exists in any regime where closure holds, and a single confirmed monopole falsifies closure. The prediction is structural and sharp, and its test is given in Section 5.

4.5 The strong interaction: the internal-closure sector

Formalization.. The strong interaction is read as the topological closure requirement of an incomplete internal braid. A colored constituent is the end of a strand that cannot terminate independently in three-dimensional space, and confinement is therefore a geometric necessity rather than a dynamical accident: the energy cost of an isolated strand end grows linearly with separation, the linear potential confirmed on the lattice, and the flux tube breaks by pair production when the stored energy reaches the threshold¹⁴. The selection of the internal symmetry group is directional and is stated at exactly that grade. Braiding governs the exchange statistics of the constituents through the braid group and its Hecke deformation, and the quantum Schur-Weyl correspondence relates that deformation to a unitary group acting on the internal space¹⁷; the rank of that group is set by the dimension of the internal fundamental and not by the number of braided strands, so the three-dimensional special unitary group is selected by taking the internal fundamental to be three-dimensional, compatible with the three observed colors, and the running of the strong coupling appears as the deformation parameter approaching the classical limit, which is asymptotic freedom^{15,16}. The choice of a three-dimensional internal space is a stipulation, not a theorem, and it is labeled a stipulation; the strand count fixes the statistics, never the rank. Baryon number is read as a configuration-level topological invariant of the color field, conserved classically by a topological argument and suppressed quantum-mechanically to a rate far below any laboratory bound^{11,13}.

The gap it fills.. The Standard Model treats confinement as a property to be demonstrated by computation and baryon-number conservation as an accidental symmetry. The gap is the absence of a geometric reason that color must be confined and that the proton must be effectively stable. The open-strand argument supplies confinement as a necessity of an unterminated strand in three dimensions, and the topological-invariant argument supplies proton stability as a conservation law rather than an accident. The group selection is honestly weaker, directional rather than unique, and is marked so.

Prediction, a geometric necessity.. Because a colored strand end cannot terminate in three dimensions, no isolated color charge and no free fractional charge exists, and because baryon number is a topological invariant, the proton does not decay at any reachable sensitivity. Both are structural and both are tested in Section 5.

4.6 The weak interaction: the screened-chiral sector

Formalization.. The weak interaction is read as a screened chiral vector configuration produced by the breaking of the electroweak symmetry. Below the breaking scale the massive mediators give the interaction its short Yukawa range, and the mass ratios among the mediators and the value of the weak mixing angle are inherited from the electroweak theory and its grand-unified normalization¹⁸. The framework reads the breaking scale as the

depth of an energetic well in the substrate's cooling history, a physical identification rather than a new mechanism, and it is important to state exactly what this reading does and does not do. It does place the breaking as a structural feature; it does not derive the value of the breaking scale, and the chirality of the interaction, that it couples only to left-handed fields, is read as a stipulation of the minimum symmetry encoding a left-right distinction rather than as a derived fact. Both the scale and the chirality are marked as unmeasured or stipulated, and neither is claimed as a result.

The gap it fills.. The Standard Model takes the breaking scale, the mediator masses, and the parity violation of the weak interaction as inputs. The gap is the absence of a structural reason for the scale and for the handedness. The framework fills this gap only partially and says so: it places the scale as a substrate feature and the handedness as a stipulation, and it does not supply the reason that would make either a necessity. This partial filling is reported at its honest grade rather than dressed as a resolution.

Prediction.. The framework's testable content in this sector is negative and structural: the hierarchy between the electroweak and the Planck scale is stabilized by the two scales being set by independent substrate features rather than by a cancellation, so no superpartner spectrum is required, and its absence at reachable collider energy is the prediction, developed in Section 5.

4.7 Elementary particles: the knot sector

Formalization.. Matter is read as topological knots of the substrate field, with mass the energy of a self-reinforcing configuration. Two configurations carry the argument. The charged lepton is read as a knotted soliton of the broken-phase direction field, whose finite three-dimensional core distributes the rest energy and removes the point-charge self-energy divergence at its geometric source, with the tree-level magnetic moment recovered and a small form-factor correction set by the core scale^{12,20}. The neutral lepton is read as a self-dual configuration: a charge-neutral fermion has vanishing gauge winding, so nothing distinguishes it from its charge conjugate, and self-duality under charge conjugation, the Majorana condition, is a topological necessity for such a configuration rather than one of two options. The neutral-fermion argument is reinforced algebraically by the real structure of the Standard Model's spectral geometry, an inherited construction that fixes the same conclusion¹⁹. The ordering of the neutral masses follows from the growth of soliton energy with topological complexity, which places the lightest state at the least complex configuration and yields normal ordering; this reading is structural and contingent on energy increasing monotonically in the relevant topological invariant, and the appropriate complexity measure is not yet fully settled, so normal ordering is reported as a structural consequence rather than a strict necessity. Two honesty marks govern this sector. The absolute mass scales of the charged and neutral leptons are not derived; the charge assignments that reproduce the mass hierarchy are fits, and the seesaw scale is a fit, both marked as such. And the

spectral-geometry input that fixes the Majorana conclusion is load-bearing and is capped at its tier rather than promoted.

The gap it fills.. The Standard Model leaves open whether neutrino mass is of Dirac or Majorana type, whether the ordering is normal or inverted, and why the point electron carries no divergent self-energy in the effective description. The gap is the absence of a structural principle selecting among these. The knot reading supplies three: the divergence is removed by a finite three-dimensional core, the Majorana type is forced for a neutral configuration, and normal ordering follows from energy growing with topological complexity, contingent on the complexity measure noted above. The absolute scales are not supplied and are marked open.

Prediction.. Because a neutral fermion configuration is self-dual, the neutrino is a Majorana particle, and because charge is a winding number, the charge spectrum is integer for color singlets and in thirds for color-wound configurations; both are geometric necessities. The normal mass ordering is a structural consequence of energy growing with topological complexity, contingent on the complexity measure noted above and not a strict necessity. Each is tested in Section 5.

4.8 The dark sector: the honest sector

Formalization.. The dark sector is read as the place where the substrate's incomplete characterization has its largest effect, and the reading is deliberately restrained. The accelerating expansion is read as a residual balanced tension of the substrate at cosmological scales, consistent with the isometric ground state of Section 4.1: the vacuum is a full balanced state and its small residual gradient sources a term of order the square of the Hubble rate, without the enormous mismatch that arises when the balanced ground state is mistaken for an empty one and its full zero-point content is counted as a gravitational source. The exact ground state cancels to zero, so the observed term is read not as a property of that exact state but as a large-scale, incompletely relaxed residual of it. This is a structural reframe of the vacuum term, and the coefficient is not derived; it is marked open. The missing-mass component is treated with greater caution still. It is reported as under-determined in the absence of a direct detection, and the framework declines to seal it. In particular, the well-known numerical proximity between the galactic acceleration scale and the product of the speed of light and the Hubble rate^{21,22} is precisely the kind of chart-manufactured coincidence that geometry before magnitude forbids as a structure fact, a target reproduced by assembling several independently unjustified factors, and it is declined as evidence here rather than used. A candidate scalar from the resolution of the strong charge-parity problem is noted as consistent with the framework, not derived from it.

The gap it fills.. The standard account supplies a cosmological constant and a dark-matter density as fitted parameters and offers no structural reason for either. The gap is the absence of a

structural reading of the vacuum term and an honest verdict on the missing mass. The framework fills the first partially, by reading the term as a residual of a balanced state, and fills the second by refusing to fill it, holding the missing-mass component under-determined pending direct detection. Declining to seal an under-determined sector is itself the discipline, and it is the sector where the discipline is most visible.

Prediction.. The one quantitative prediction the framework will stand behind here is a structural consequence of the residual-tension reading: the equation of state stays at or above minus one and departs from minus one by a small positive amount today as the residual relaxes, with no crossing below minus one at any redshift. This is the single prediction in the paper that currently runs against its author. Present baryon-acoustic-oscillation data prefer an evolving dark energy whose value today lies above minus one, which agrees with the framework, but whose recent past lies below minus one, a phantom excursion the framework forbids. The framework is therefore disfavored on the time-evolution while consistent on the present sign, and a fixed cosmological constant is disfavored as well. It is tested in Section 5, and it is reported without softening.

4.9 The black hole: the horizon sector

Formalization.. The black hole is read as a horizon configuration, and the reading is thin by honesty because the framework authors no interior theory. What is placed is precise: the horizon is the thermodynamic surface on which the heat-area proportionality of Section 4.3 is imposed to yield the field equations, so the black hole is exactly where the equation-of-state reading of gravity is anchored, and the entropy assigned to the horizon is the inherited area law extended by the holographic bound^{28,29}. The interior, the singularity, and the information question are not placed and are marked open: an incompletely characterized substrate supplies no interior structure, no resolution of the singularity, and no resolution of the information question, and the substrate's single unmeasured input, developed next, becomes in this sector a horizon on the theory itself.

The gap it fills.. General relativity predicts a singular interior and standard semiclassical gravity leaves the information question unresolved. The gap is a structural account of the interior. The framework does not fill this gap and states so plainly; it fills only the horizon, as the surface where the thermodynamic reading of gravity is anchored, and it inherits the entropy law rather than deriving it. The honest report of an unfilled gap is preferred here to an invented interior.

Prediction, a geometric necessity.. Because the mediator is spin two, the ringdown of a horizon carries only quadrupolar radiation with no scalar or vector component, which is the same structural necessity as in Section 4.3 read at the horizon, and it is tested in Section 5.

4.10 The single unmeasured input

Formalization.. The sectors above share one gap, and naming it precisely is the unifying content of the framework. The substrate is posited but not characterized: it is assigned no temperature, no fluctuation spectrum, no coupling, and no elementary length or time. This single omission is where every quantitative constant not inherited from established physics resides. A response coefficient is fixed by a fluctuation-dissipation relation that requires the medium's temperature, its fluctuation spectrum, and its coupling, and the substrate supplies none of the three; that is precisely why the galactic acceleration scale, the electromagnetic response coefficients and the speed of light, the absolute lepton masses, the charge assignments that reproduce the mass hierarchy, the electroweak breaking scale, and the seesaw scale are each a family of allowed values or an open target rather than a computed number. The constraints of the framework fix the class of each and never the member. This continuous input is one thing, and the framework carries beside it a small closed set of discrete structural stipulations, the three-dimensional internal space of the strong sector and the left-handed chirality of the weak sector, marked as stipulations in the sectors above and not claimed to reduce to the substrate characterization. The honest statement is therefore one continuous unmeasured input plus a fixed and disclosed set of discrete stipulations, not a single input that also fixes the discrete structure.

The sectors therefore cohere not because one root derives the forces, but because their quantitative predictions all fail in the same nameable place and their structures are all read in the same order. This is a structural and organizational unification, not a computational one, and the distinction is the whole of its honesty. What would close the gap is a characterized substrate, its temperature, fluctuation spectrum, coupling, and elementary scale measured or derived rather than posited, and that is an experimental program with a definite target, not a philosophical remark. The framework's discipline is to state this price in advance rather than to hide it inside a fitted coincidence.

5. Falsifiable predictions

The predictions below are constructed so that their confirmation criteria are geometric necessities or structural consequences rather than fitted numbers. Each is stated as a phenomenon with a threshold, a method of confirmation with named instruments, an expected outcome, and a null result that would falsify the associated structure. Each is testable by more than one instrument using an orthogonal modality.

Box 1 | Prediction one · gravitational radiation is purely quadrupolar

The prediction.. Gravitational waves carry only the two transverse traceless quadrupolar polarizations. No scalar breathing polarization and no vector polarization is present at any amplitude. The threshold is a null on the non-tensor modes at five standard deviations.

Method of confirmation.. Full polarization decomposition of compact-binary signals by a network with independent baselines. The next-generation ground detectors, the Einstein Telescope and Cosmic Explorer, resolve the polarization content directly; the present network of advanced detectors constrains it from event populations²⁵.

Expected outcome.. Every signal consistent with pure quadrupolar content, with the scalar and vector amplitudes consistent with zero.

Null hypothesis.. Detection of a scalar or vector polarization component at five standard deviations, confirmed by independent waveform analysis, falsifies the rank-two source and the spin-two exhaustion of Section 4.3.

Box 2 | Prediction two · no magnetic monopole

The prediction.. No magnetic monopole exists in any accessible regime. The threshold is the continued null across all searched channels.

Method of confirmation.. Direct collider and cosmic searches by orthogonal modalities, the trapping and induction detectors at accelerators and the large-volume neutrino observatories searching for cosmic monopoles.

Expected outcome.. Continued absence of any monopole signature at all searched channels and energies.

Null hypothesis.. A single confirmed magnetic monopole, at any energy, falsifies the closure axiom of Section 4.4 and with it the structural exclusion of the co-exact sector.

Box 3 | Prediction three · exact color confinement and no free fractional charge

The prediction.. No isolated color charge and no free fractional electric charge is ever observed. The threshold is the continued null on free-quark and fractional-charge searches at five standard deviations.

Method of confirmation.. Millikan-type fractional-charge searches in bulk matter and the jet and hadronization structure of high-energy collisions, two orthogonal modalities.

Expected outcome.. Confinement exact at all separations, every colored object appearing only in neutral combinations, and no fractional residual charge in bulk matter.

Null hypothesis.. Observation of a free quark or a stable isolated fractional charge falsifies the open-strand geometric necessity of Section 4.5.

Box 4 | Prediction four · proton effective stability

The prediction.. The proton does not decay at any reachable sensitivity, its lifetime lying far above the ranges predicted by grand unified theories. The threshold is a strict null in every kinematically allowed channel at five standard deviations.

Method of confirmation.. The next-generation underground detectors by orthogonal detection media, the large water Cherenkov detector Hyper-Kamiokande and the liquid-argon detector DUNE, together with the present bounds from Super-Kamiokande³¹.

Expected outcome.. Continued null in all channels through the operational lifetime of the detectors.

Null hypothesis.. A single confirmed proton-decay event, surviving background discrimination at five standard deviations, falsifies the topological-invariant conservation of baryon number of Section 4.5.

Box 5 | Prediction five · normal neutrino ordering and Majorana character

The prediction.. The neutrino mass ordering is normal, and the neutrino is its own antiparticle. The thresholds are a determination of normal ordering at five standard deviations and a detection of neutrinoless double beta decay consistent across two isotopes.

Method of confirmation.. The medium-baseline reactor experiment JUNO for the ordering by spectral distortion³², cross-checked by atmospheric measurements; and the neutrinoless double-beta-decay experiments in xenon and germanium for the Majorana character, two orthogonal isotopes.

Expected outcome.. Normal ordering established at high significance, and, if the neutrino is Majorana, a neutrinoless double-beta-decay signal whose inferred effective mass agrees across isotopes. Under normal ordering the effective Majorana mass can be driven near zero by a cancellation among the mass eigenstates for certain values of the Majorana phases, so a positive signal is decisive while a null is not.

Null hypothesis.. A confirmed inverted ordering falsifies the energy-complexity reading of Section 4.7. For the Majorana character, a positive neutrinoless-double-beta-decay signal confirms it, and a determination that the neutrino is Dirac, by a lepton-number-conserving measurement, falsifies the self-duality necessity for a neutral configuration; a null neutrinoless-double-beta-decay result alone does not falsify, because the effective mass can vanish under normal ordering. A definite effective-mass band, on the order of a few to several millielectronvolts, is predicted only under an added stipulation fixing the Majorana phases, and a null below that band would then falsify.

Box 6 | Prediction six · no fourth fermion generation

The prediction.. No fourth generation of quarks or leptons exists at any energy. This is reported as an inherited empirical result read through a structural conjecture, not as a geometric necessity: the measured count of three light neutrino species from the width of the neutral weak boson and the absence of a fourth generation at collider energies are the empirical basis, and the framework reads the count as a cap on topological complexity without deriving the number three, which remains an open structural problem.

Method of confirmation.. Direct searches at the high-luminosity collider and the precision count of light neutrino species from the width of the neutral weak boson³⁴, two orthogonal modalities.

Expected outcome.. Continued absence of a fourth-generation state and a light-neutrino count consistent with three.

Null hypothesis.. Detection of a sequential fourth-generation fermion, or a determination of the light-neutrino count above three at five standard deviations, falsifies the conjecture that topological complexity is capped at the third generation; it does not touch the inherited count, which the measurement itself fixes.

Box 7 | Prediction seven · the dark-energy equation of state departs from minus one

The prediction.. The equation of state stays at or above minus one at every redshift, with a small positive departure from minus one today as the residual substrate tension relaxes, and no crossing below minus one in the past. This is the one prediction that currently runs against the author. The threshold is a present-day value above minus one together with the absence of any phantom crossing.

Method of confirmation.. The joint analysis of baryon-acoustic-oscillation surveys, weak-lensing surveys, and cosmic-microwave-background lensing by orthogonal cosmological probes, the galaxy survey DESI, the space mission Euclid, and the next-generation microwave-background array^{30,33}.

Expected outcome.. A present-day value above minus one, which current baryon-acoustic-oscillation data support, together with no excursion below minus one at any redshift. Current data instead prefer an evolving dark energy that dips below minus one in the recent past and crosses back near redshift one-half, rejecting a fixed cosmological constant at roughly three to four standard deviations depending on the supernova sample; the present sign agrees with the framework and the past phantom excursion does not.

Null hypothesis.. A confirmed crossing below minus one at any redshift, or a value fixed at minus one to high precision across all bins, falsifies the residual-tension reading of Section 4.8. The first of these is the excursion current data already mildly prefer, so this null is live, and the framework reports itself as currently disfavored on the time-evolution.

The predictions are dominated by geometric necessity: the purely quadrupolar radiation, the excluded monopole, the exact confinement, and the Majorana character are each forced by a topology or a source structure rather than by a fitted value. The effective proton stability follows as a structural consequence of a conserved topological invariant, and the normal ordering as a structural consequence of energy growing with topological complexity. Two predictions carry weaker grades and are marked as such: the generation cap is an inherited count read through an open structural conjecture rather than a necessity, and the dark-energy departure carries only the sign of a quantitative magnitude and is currently disfavored on its time-evolution. A framework whose empirical content is mostly necessity is falsified cleanly: any one confirmed counterexample removes the structure that entails it, with no free parameter available to absorb the failure.

6. Discussion

The account reframes several adjacent problems by placing them on one substrate. The hierarchy between the electroweak and the Planck scale becomes a statement that two independent substrate features set two independent scales, so that the quadratic sensitivity of the lighter scale to the heavier is an artifact of a

regularization that couples sectors with no physical coupling; this is a structural reframe and is labeled one, not a computation of the ratio. The cosmological-constant mismatch becomes a statement that the balanced ground state was mistaken for an empty one, and its resolution is likewise structural and awaits the coefficient that only a characterized substrate would supply. The generation structure is read as a topological-complexity cap rather than an unexplained triplication, though the framework does not derive the number three and the bounding invariant remains open, so this reading is a structural conjecture and not a result. In each case the reframe is reported at structural grade, and the quantitative content is explicitly deferred to the single unmeasured input.

Several objections deserve direct engagement. The first is that the group selection for the strong interaction is not a uniqueness theorem, and this is granted without reservation: the argument selects the three-dimensional special unitary group by stipulating a three-dimensional internal space compatible with the three observed colors, but it does not exclude every alternative product structure, and that internal dimension is a stipulation rather than a derived fact. The second is that the framework imports established results, the spin-two exhaustion, the Hodge decomposition, the division-algebra classification, the spectral geometry, and rests weight on them; this is granted and is the reason each such result is cited and capped at its tier, with the framework claiming only the arrangement and never the imported theorem. The third is that the framework predicts no new numerical constant; this is not an objection but the central honest content, and it is stated as the single unmeasured input rather than hidden. The fourth is that a structural reframe is not a derivation; this too is granted, and the paper is careful never to write that a problem is resolved where only its framing has changed.

The limitations are therefore explicit and are the same limitation seen from different sectors. The absolute masses, the electroweak scale, the electromagnetic response coefficients and the speed of light, the seesaw scale, the dark-energy coefficient, and the missing-mass microphysics are not derived, and the black-hole interior is not placed. All of these reside in the substrate's missing thermodynamic characterization, and the framework's contribution is to locate them in one place and to state the measurement that would address them, not to compute them. What the framework offers in exchange is a set of predictions that are mostly geometric necessity and therefore cleanly falsifiable, and a discipline that declines the fitted coincidences that would make the account appear to derive more than it does.

7. Conclusion

The interactions and the matter spectrum are described by the standard theories without their structure being derived, and the gaps, the gauge group, the force hierarchy, the spin content, the dark sector, and the vacuum term, are structural rather than computational. This paper has placed the four forces and the particle spectrum as configurations of one actuating continuous substrate, read through a three-way orthogonal decomposition and

in the order geometry before magnitude, with each force a distinct source-tensor configuration and each particle a topological knot. The empirical content is deliberately dominated by geometric necessity: gravitational radiation purely quadrupolar, magnetic monopoles absent, color exactly confined, and the neutrino a Majorana particle, with the proton's effective stability and the normal mass ordering following as structural consequences, the three-generation count inherited and read through an open conjecture, and the dark-energy equation of state held at or above minus one with a small present departure that current data disfavor.

The primary near-term test is the polarization content of gravitational radiation, a clean structural necessity resolvable by the next-generation detector network, and the one prediction that currently runs against the framework is the dark-energy equation of state, reported without softening. The experimental community is asked to resolve the gravitational-wave polarization content, to continue the monopole and free-charge and proton-decay searches whose nulls confirm the structural exclusions, to fix the neutrino ordering and character, and to determine whether the dark-energy equation of state departs from minus one. The single most important open question is the thermodynamic characterization of the substrate, the temperature, fluctuation spectrum, coupling, and elementary scale in which every remaining continuous constant resides, beside the small disclosed set of discrete structural stipulations the framework marks rather than derives. Acceptance of this account would entail reading the four forces and matter as configurations of one substrate placed by geometry rather than as separate structures fixed by measurement, with the continuous constants of nature deferred, openly, to a single unmeasured input, and the residual discrete structure named as a closed set of stipulations rather than hidden.

Appendix A: Foundational principles

The following principles are presented as standalone physical statements, each independently motivated and independently testable within the native discipline.

Principle one, the actuation floor.. Every existent carries positive kinetic content at the level of its constituents. For confined quantum existents this is the strictly positive kinetic-energy bound from the canonical commutation relation, the strictly positive zero-point energy, and the rigorous third law forbidding a state at exact zero; its extension to every existent is a premise. Function in the argument: it makes the substrate active rather than inert and forbids an empty ground state.

Principle two, the balanced ground state.. The ground state of the substrate has vanishing vector sum of its internal components and nonzero magnitude of each component. Vector cancellation is not scalar absence. Function: it grounds the reading of the cosmological term as a residual of a balanced state rather than the energy of an empty one.

Principle three, three orthogonal warrants.. A complete claim about an actuating configuration resolves into a formal-structural, a thermodynamic, and a registrational warrant, mutually independent, the count of three forced semantically by a deletion test and algebraically by the classification of the real division algebras. Function: it requires every sector to be described formally, energetically, and observationally at once, and it fixes the sign of a structure to be read from its geometry rather than its magnitude.

Principle four, geometry before magnitude.. The formal and geometric warrants fix what a configuration is and are read first; the numerical warrant reports magnitude only and is read last, and a magnitude manufactured by a choice of chart, carrying no structural information, may not enter a verdict as a structure fact. Function: it disqualifies fitted coincidences and enforces that predictions be structural necessities.

Principle five, mass as topological energy.. Localized mass is the energy of a self-reinforcing topological configuration of the continuous field, and conservation laws governing particle stability are topological invariants of that configuration rather than externally imposed quantum numbers. Function: it makes the electron core finite, forces the neutral fermion to be self-dual, and reads baryon number as a conserved invariant.

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full at the reference below.

References

1. Fierz, M. and Pauli, W. 1939. On relativistic wave equations for particles of arbitrary spin in an electromagnetic field. *Proceedings of the Royal Society A* 173, 211.
2. Wigner, E. 1939. On unitary representations of the inhomogeneous Lorentz group. *Annals of Mathematics* 40, 149.
3. Jacobson, T. 1995. Thermodynamics of spacetime: the Einstein equation of state. *Physical Review Letters* 75, 1260.
4. Verlinde, E. 2011. On the origin of gravity and the laws of Newton. *Journal of High Energy Physics* 2011(4), 29.
5. Frobenius, G. 1878. Über lineare Substitutionen und bilineare Formen. *Journal für die reine und angewandte Mathematik* 84, 1.
6. Hurwitz, A. 1898. Über die Composition der quadratischen Formen von beliebig vielen Variablen. *Nachrichten der Gesellschaft der Wissenschaften zu Göttingen*, 309.
7. Bott, R. and Milnor, J. 1958. On the parallelizability of the spheres. *Bulletin of the American Mathematical Society* 64, 87.
8. Friedrichs, K. O. 1955. Differential forms on Riemannian manifolds. *Communications on Pure and Applied Mathematics* 8, 551.
9. Aharonov, Y. and Bohm, D. 1959. Significance of electromagnetic potentials in the quantum theory. *Physical Review* 115, 485.
10. Dirac, P. A. M. 1931. Quantised singularities in the electromagnetic field. *Proceedings of the Royal Society A* 133, 60.
11. Faddeev, L. and Niemi, A. J. 1997. Stable knot-like structures in classical field theory. *Nature* 387, 58.
12. Batty, R. A. and Sutcliffe, P. M. 1998. Knots as stable soliton solutions in a three-dimensional classical field theory. *Physical Review Letters* 81, 4798.

13. Cho, Y. M. 1980. A restricted gauge theory. *Physical Review D* 21, 1080.
14. Skyrme, T. H. R. 1962. A unified field theory of mesons and baryons. *Nuclear Physics* 31, 556.
15. Gross, D. J. and Wilczek, F. 1973. Ultraviolet behavior of non-abelian gauge theories. *Physical Review Letters* 30, 1343.
16. Politzer, H. D. 1973. Reliable perturbative results for strong interactions. *Physical Review Letters* 30, 1346.
17. Jimbo, M. 1986. A q-difference analogue of $U(g)$ and the Yang-Baxter equation. *Letters in Mathematical Physics* 10, 63.
18. Georgi, H. and Glashow, S. L. 1974. Unity of all elementary particle forces. *Physical Review Letters* 32, 438.
19. Chamseddine, A. H. and Connes, A. 2007. Why the standard model. *Journal of Geometry and Physics* 58, 38.
20. Finkelstein, D. and Rubinstein, J. 1968. Connection between spin, statistics, and kinks. *Journal of Mathematical Physics* 9, 1762.
21. Milgrom, M. 1983. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *The Astrophysical Journal* 270, 365.
22. McGaugh, S. S., Lelli, F. and Schombert, J. M. 2016. Radial acceleration relation in rotationally supported galaxies. *Physical Review Letters* 117, 201101.
23. MICROSCOPE Collaboration. 2022. MICROSCOPE mission: final results of the test of the equivalence principle. *Physical Review Letters* 129, 121102.
24. LIGO Scientific and Virgo Collaborations. 2016. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters* 116, 061102.
25. LIGO-Virgo-KAGRA Collaboration. 2021. Tests of general relativity with binary black hole signals. *Physical Review D* 103, 122002.
26. Masanes, L. and Oppenheim, J. 2017. A general derivation and quantification of the third law of thermodynamics. *Nature Communications* 8, 14538.
27. Landauer, R. 1961. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development* 5, 183.
28. Bekenstein, J. D. 1973. Black holes and entropy. *Physical Review D* 7, 2333.
29. Bousso, R. 1999. A covariant entropy conjecture. *Journal of High Energy Physics* 1999(07), 004.
30. DESI Collaboration. 2024. DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations. *arXiv:2404.03002*.
31. Super-Kamiokande Collaboration (Takenaka, A. et al.). 2020. Search for proton decay via two-body modes with an enlarged fiducial volume in Super-Kamiokande. *Physical Review D* 102, 112011.
32. JUNO Collaboration (An, F. et al.). 2016. Neutrino physics with JUNO. *Journal of Physics G* 43, 030401.
33. Planck Collaboration. 2020. Planck 2018 results VI: cosmological parameters. *Astronomy and Astrophysics* 641, A6.
34. ALEPH, DELPHI, L3, OPAL, and SLD Collaborations. 2006. Precision electroweak measurements on the Z resonance. *Physics Reports* 427, 257.
35. Islam, M. TRISDUCTION: A Linguistically, Topologically, and Mathematically Sealed Verification Architecture. Triaxial Orthogonality, Twelve-Gate Closure, the Quaternionic Completion, the Root Axiom, and the Master Pre-Sealed Proposition Ledger. Zenodo, version 4, 19 June 2026. DOI 10.5281/zenodo.20757507. <https://zenodo.org/records/20757507>. Mirror: PhilArchive record ISLTTG, <https://philpapers.org/rec/ISLTTG>. Master reference, continuously updated at the same location.

Competing interests

The author declares no competing interests.

Correspondence

Correspondence and requests for materials should be addressed to islammm@alumni.iu.edu.